

FEASIBILITY MEMO

To: Mercer ED Board

From: Vishwesh Pattanaik, Clinical AI and Innovation Unit

Re: Feasibility of the de-identified ED arrivals dataset for triage modelling

Dataset: yaleemmlc_admissionprediction_triage.csv (55,121 patients, 226 columns)

Verdict

The dataset is large, complete and clinically rich, and it is good enough to build a baseline triage model, provided its single-site, pre-processed nature and its class imbalance are documented and respected.

Dataset summary

The dataset holds 55,121 emergency department arrivals from a single US academic hospital. It has 226 columns: 26 core fields covering demographics, triage vital signs, arrival details and disposition, and 200 one-hot flags for chief complaints. The triage target is the Emergency Severity Index (ESI), scored from 1 (most urgent) to 5 (least). The target is complete for every patient. The middle band, ESI 3, is 49% of all arrivals, so the data is concentrated exactly where this project is aimed.

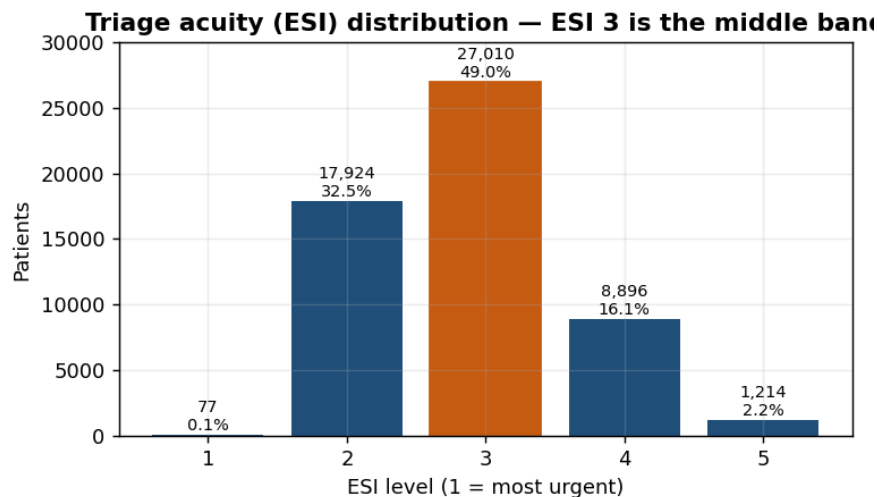


Figure 1. Triage acuity distribution. The middle band (ESI 3) is 49% of arrivals.

Top three quality concerns

1. The data is pre-processed, not raw. Every column is 100% complete and no vital sign falls outside a plausible range. Real triage data is never this clean. This is the processed release of the source dataset: vital signs were imputed and complaints were one-hot-encoded before we received it. We cannot audit the original missingness or the imputation choices, and this cleanliness will not carry over to Mercer's own EHR extract, which will need its own data-quality handling.

Missingness summary (df.isnull().sum())

column	dtype	n_missing	pct_missing	range / unique
dep_name	object	0	0.0	3 unique
esi	float64	0	0.0	1 to 5
age	float64	0	0.0	18 to 107
gender	object	0	0.0	2 unique
ethnicity	object	0	0.0	4 unique
race	object	0	0.0	8 unique
lang	object	0	0.0	2 unique
religion	object	0	0.0	12 unique
maritalstatus	object	0	0.0	10 unique
employmentstatus	object	0	0.0	10 unique
insurance_status	object	0	0.0	5 unique
disposition	object	0	0.0	2 unique
arrivalmode	object	0	0.0	7 unique
arrivalmonth	object	0	0.0	12 unique
arrivalday	object	0	0.0	7 unique
arrivalhour_bin	object	0	0.0	6 unique
previousdispo	object	0	0.0	10 unique
triage_vital_hr	float64	0	0.0	32 to 221
triage_vital_sbp	float64	0	0.0	53 to 266
triage_vital_dbp	float64	0	0.0	27 to 189
triage_vital_rr	float64	0	0.0	8 to 66
triage_vital_o2	float64	0	0.0	62 to 99
triage_vital_o2_device	float64	0	0.0	0 to 1
triage_vital_temp	float64	0	0.0	91.5 to 106
triage_glucose	float64	0	0.0	16 to 1066
cc_* (200 cols)	int64	0	0.0	0/1

Figure 2. Missingness summary. Zero missing cells across all 226 columns.

2. Severe class imbalance. ESI 1 has only 77 patients and ESI 5 only 1,214. Reliable prediction of these extremes is unlikely, and headline accuracy would be misleading. Performance must be reported per acuity level, not as a single overall figure.

3. Sparse chief-complaint features. Of the 200 complaint flags, 49 have fewer than 50 positive cases. Left unmanaged, these very rare features invite overfitting and should be grouped or dropped before modelling.

Top three reasons to proceed

1. Scale and completeness. 55,121 labelled patients with no missing target is more than enough for a credible baseline model.

2. The target matches the project. The middle band is nearly half the data, so there is ample signal exactly where the project intends to work.

3. Clinical face validity. The features relate to acuity in the directions a clinician would expect. Chest pain, shortness of breath and altered mental status pull toward higher acuity; sore throat and dental pain toward lower. This is good evidence that the data behaves sensibly.

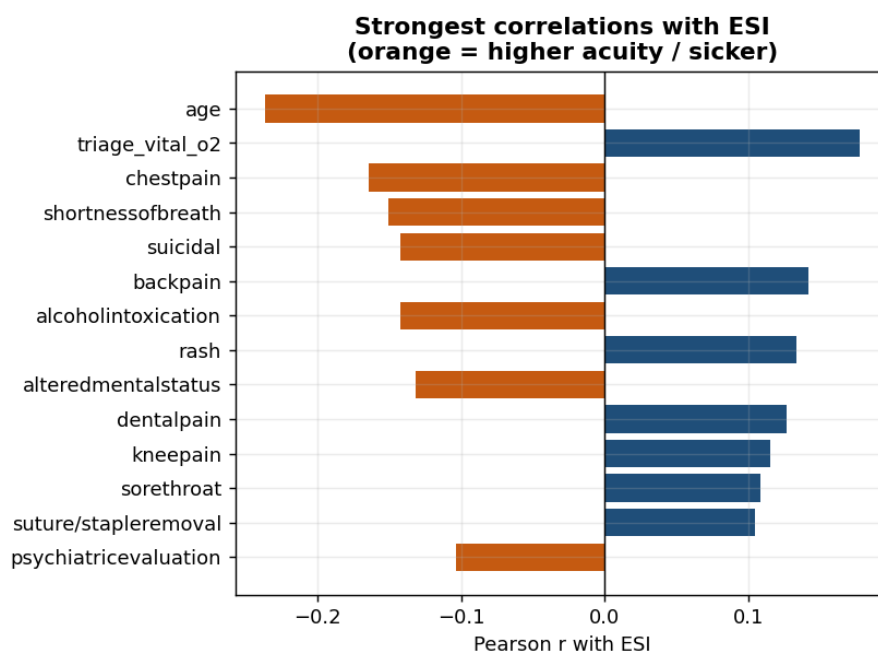


Figure 3. Strongest correlations with ESI. Orange indicates higher acuity (a sicker patient).

Caveats

Four caveats should travel with any model built on this data. First, it comes from a single US academic ED; the case-mix and access patterns differ from Mercer and from a Caribbean population, so a model trained here would need local re-validation before any transfer. Second, no single feature is strongly predictive (the largest correlation, age, is about 0.24), so the signal is spread across many variables, and a simple rule will not suffice. Third, the imputation and encoding decisions are inherited and cannot be inspected. Fourth, de-identified is not risk-free. The file is kept out of version control for its size and sensitivity, and only summary statistics leave the analysis environment.

Top-10 predictive feature shortlist

Ranked by clinical judgement together with simple correlation against ESI. Each is justified in one line.

Rank	Feature	Why it matters
1	age	Older patients decompensate faster and show the strongest single correlation with acuity (r about -0.24).
2	SpO2	Low oxygen saturation is an immediate, objective red flag for a high-acuity patient.
3	Respiratory rate	An early and sensitive sign of deterioration, often changing before other vitals.
4	Heart rate	Tachycardia signals shock, sepsis or severe distress.
5	Systolic BP	Hypotension defines instability even when the patient otherwise looks stable.
6	Chest pain	Carries a high pre-test probability of a life-threatening cause (r about -0.16).
7	Shortness of breath	Maps directly to airway and breathing priority at triage (r about -0.15).
8	Altered mental status	A recognised top-tier trigger for urgent assessment (r about -0.13).
9	Glucose	Extreme values flag diabetic emergencies such as DKA or hypoglycaemia.
10	Arrival mode	Ambulance arrival is a strong proxy for pre-hospital acuity.